🛡️ Malware Analyzer — User Guide

🔍 Overview

This tool is a lightweight malware analysis utility designed to inspect executable files (.exe) for suspicious behavior.

It combines multiple analysis techniques into one interface:

- Hashing
- String analysis
- PE file inspection
- YARA rule detection

It also generates a detailed **PDF report** for further review.

🚀 Features

- 🔑 **File Hashing**
  - Generates MD5 and SHA256 fingerprints
- 🧠 **Suspicious String Detection**
  - Detects keywords like:
    - powershell
    - http
    - cmd
    - .onion
    - keylogger
- 🧩 **PE File Analysis**
  - Extracts imported DLLs from executables
- 🧬 **YARA Rule Matching**
  - Detects known malicious patterns using custom rules
- 📄 **Automatic Report Generation**
  - Creates a structured PDF report after analysis

- 🖥️ **GUI Interface**
  - Simple and user-friendly (Tkinter-based)

📥 Getting Started

1. Install Python Dependencies

pip install pefile yara-python colorama fpdf

2. Run the Tool

python your_file_name.py

🎯 How to Use

1. Launch the application
2. Click **"🔍 Select File & Analyze"**
3. Choose an .exe file
4. Wait for analysis to complete
5. View results inside the window
6. A **PDF report** will be generated automatically

🔍 What the Tool Actually Does

🔑 Hash Analysis

- Generates:
  - MD5
  - SHA256
- Useful for identifying known malware samples

String Analysis

- Extracts readable strings from file
- Flags suspicious keywords like:
  - network calls
  - command execution
  - scripting usage

PE File Analysis

- Checks imported libraries (DLLs)
- Helps detect:
  - networking behavior
  - system-level access

YARA Scanning

Uses rule-based detection to flag patterns like:

- Downloaders
- Command execution behavior
- Web requests

Output

After scanning:

- Results are displayed in the GUI
- A PDF report is generated:

filename_report.pdf

Contains:

- Hash values
- Suspicious strings

- Imports
- YARA matches

⚠ Important Notes

- This tool is for **educational and basic analysis only**
- It is **not a full antivirus solution**
- Some malware may bypass detection
- Requires internet/tools setup (like strings command)

🛠 Troubleshooting

❌ "strings command not found"

- On Windows, install Sysinternals strings.exe
- Add it to PATH

❌ YARA errors

- Make sure yara-python is installed correctly

❌ Permission issues

- Run the program as **Administrator**

❌ No output or crash

- Check if selected file is valid .exe

🎯 Use Cases

- Malware analysis practice
- Cybersecurity learning
- Suspicious file inspection
- Small business security checks

🧠 Security Insight

If a file shows:

- YARA matches ❌
- Suspicious strings ❌

👉 Treat it as **potentially dangerous**

🚀 Final Note

This tool doesn't just scan files...

It teaches you how malware *whispers before it attacks* —
through strings, imports, and silent patterns hidden beneath the surface.